CODE//CONNECT'26

VIBE CODING HACKATHON

Problem Statements: Three Real-World Challenges

Technology is most valuable when it solves problems that matter. Below are the three problem statements for CODE//CONNECT'26. Each describes a real situation, the core challenge to solve, and the questions that should guide your thinking. The statements belong to three unrelated domains, so every team chooses one and builds something that makes a real difference. Teams define their own slice of the problem and are free to choose any platform, data source or approach.

PROBLEMS AT A GLANCE

#	Domain	Focus	Core question
01	Disaster Response	Finding who needs help most	How do we identify the most at-risk people and match scarce resources to them before it is critical?
02	Threat Prevention	Acting on threats before harm happens	How do we detect a developing threat and respond early enough to prevent or reduce harm?
03	Misinformation Control	Slowing false news while protecting privacy	How do we slow misinformation in encrypted groups without censorship or loss of privacy?

GROUND RULES FOR ALL TEAMS

- Choose one problem statement and complete it.
- Deliver a working prototype that can be shown live, not only slides.
- Explain your trade-offs: what you chose, what you left out and why.
- Be explicit about how your solution behaves when data is incomplete or wrong.
- Design for real users, real constraints and real contexts.

BUILD IT. SHIP IT. MAKE IT MATTER.

01 Disaster Response

Focus: *finding who needs help most*

Problem number	01
Domain	Disaster response and emergency coordination
Focus	Finding who needs help most

THE PROBLEM

During floods, heatwaves and other disasters, warnings may not reach everyone. Elderly residents, people living alone, low-income neighbourhoods, people with disabilities and those without smartphones or reliable signal are the least likely to receive alerts and the most likely to be in danger. At the same time, rescue teams, vehicles, shelters and medical supplies are limited, so every decision about who to help first carries a cost.

THE CHALLENGE

CORE CHALLENGE

How can technology identify who needs help most and coordinate available resources to reach them before the situation becomes critical?

KEY QUESTIONS TO THINK ABOUT

Who is most at risk?	Vulnerability depends on location, age, health, mobility, housing and access to information. Teams must decide what signals reveal risk and how to find people when warnings do not reach everyone.
Who gets help first?	Rescue teams, vehicles, shelters and medical supplies are finite. The system must match scarce resources to the greatest need, and justify its priority order in a way responders can trust.
How early can you act?	Help is most effective before a situation becomes critical. Think about early signals, forecasts, crowd reports and pre-positioning of resources rather than reacting after the fact.

WHO IS AFFECTED

- Residents of flood-prone and low-lying areas with no early warning
- Elderly, disabled, chronically ill or isolated people who cannot self-evacuate
- Rescue coordinators and volunteers with limited vehicles and supplies
- Shelter and relief managers who must balance capacity against demand

WHERE IT GETS HARD

- **Incomplete information:** reports will be late, duplicated, inaccurate or missing entirely from some areas.
- **Competing priorities:** a medical emergency, a stranded family and a flooded road all demand the same team.
- **Connectivity failure:** networks and power often fail exactly when the system is needed.

02 Threat Prevention

Focus: *acting on threats before harm happens*

Problem number	02
Domain	Public safety and early threat detection
Focus	Acting on threats before harm happens

THE PROBLEM

Cameras and sensors can detect and record incidents such as theft, violence, accidents, fires and other dangerous situations. But in most deployments the footage is only reviewed after the event, or a human operator watching many screens misses what is developing. By the time anyone notices and responds, harm may already have occurred.

THE CHALLENGE

CORE CHALLENGE

How can technology detect a developing threat and take an appropriate action early enough to prevent or reduce harm?

KEY QUESTIONS TO THINK ABOUT

Detecting early	What signs show a threat is developing, not just that it has already happened? Consider behaviour patterns, escalation, crowding, smoke, abandoned objects or unusual movement.
Responding right	What action fits the situation, and who needs to be alerted? A warning, a call to security, an emergency service dispatch and a local alarm are very different responses.
Acting in time	How do you respond fast enough to make a real difference? Detection that arrives after the incident has no preventive value.

WHO IS AFFECTED

- Shop owners, markets, schools, hospitals and residential communities with camera systems
- Security staff who cannot watch every feed continuously
- Emergency responders who need accurate, early and actionable alerts
- Ordinary people whose privacy and dignity must be protected by the system

WHERE IT GETS HARD

- **False alarms:** too many wrong alerts cause people to ignore the system, while missed threats cost lives.
- **Proportionality:** the response must match the severity; escalating too aggressively causes its own harm.
- **Privacy and bias:** surveillance can intrude on people and may unfairly flag certain groups.
- **Limited hardware:** many real sites have low-quality cameras, poor lighting and weak connectivity.

03 Misinformation Control

Focus: *slowing false news while protecting privacy*

Problem number	03
Domain	Misinformation, trust and digital privacy
Focus	Slowing false news while protecting privacy

THE PROBLEM

During elections and health crises, false information spreads rapidly through WhatsApp groups. These groups are end-to-end encrypted, so their content cannot be seen by platforms, researchers or authorities. This makes conventional moderation and fact-checking impossible, while rumours reach millions of people within hours.

THE CHALLENGE

> **CORE CHALLENGE**
> **Design a system that slows the spread of misinformation without censorship, while keeping user privacy intact.**

KEY QUESTIONS TO THINK ABOUT

Privacy vs. safety	Protecting privacy limits what any system can see or do. Teams must find interventions that work without reading private messages or breaking encryption.
Adversarial users	Some people will actively try to get around the solution. A credible design anticipates evasion, abuse and gaming.
Culture and language	Every community has its own context, languages and trusted voices. One approach will not fit all, so the design must adapt locally.
Defining truth	Deciding what counts as false is itself a hard problem. The system must avoid becoming an arbiter of truth or a tool for censorship.

WHO IS AFFECTED

- Everyday WhatsApp users who forward messages without verifying them
- Community admins who manage large groups and want to reduce harm
- Fact-checkers and health or election bodies who cannot reach private networks
- Citizens of every language community, including low-literacy users

WHERE IT GETS HARD

- **Encryption:** the system cannot rely on reading, scanning or storing message content.
- **Free speech:** any intervention must slow spread, not remove or silence legitimate opinion.
- **Manipulation:** bad actors may exploit reporting, labelling or friction mechanisms themselves.
- **Local context:** satire, religion, politics and slang differ across languages and regions.